Question 3

```
clearvars, clc

g_st = readmatrix('q3dat.xlsx')
```

```
g_st = 4×4
    0.7742   -0.4195    0.4739   -0.7525
   -0.1561    0.5991    0.7853    1.0747
   -0.6134   -0.6820    0.3983    0.1726
         0         0         0    1.0000
```

```
g_st_alt = [0.77421181 -0.41948725 0.47394780 -0.75250222;
-0.15613988 0.59908988 0.78530991 1.07468455;
-0.61336482 -0.68199836 0.39832377 0.17260115;
0.00000000 0.00000000 0.00000000 1.00000000]
```

```
g_st_alt = 4×4
    0.7742   -0.4195    0.4739   -0.7525
   -0.1561    0.5991    0.7853    1.0747
   -0.6134   -0.6820    0.3983    0.1726
         0         0         0    1.0000
```

```
shouldbezero4by4_single_precision = g_st - g_st_alt
```

```
shouldbezero4by4_single_precision = 4×4
10^-8 ×
    0.2837    0.0113   -0.1138   -0.4002
    0.3068   -0.0220    0.0021    0.1277
   -0.4306   -0.0062    0.4305   -0.2044
         0         0         0         0
```

```
l_0 = 0.65
```

```
l_0 =
0.6500
```

```
l_1 = 0.8
```

```
l_1 =
0.8000
```

```
l_2 = 0.5
```

```
l_2 =
0.5000
```

```
l_3 = 0.2
```

```
l_3 =
0.2000
```

```
g_st0 = [eye(3),[0;l_1+l_2+l_3;l_0];zeros(1,3),1]
```

```
g_st0 = 4×4
    1.0000         0         0         0
         0    1.0000         0    1.5000
         0         0    1.0000    0.6500
```

```
         0         0         0    1.0000
```

```
pb = [0;0;l_0];
q2 = [0;l_1;l_0];
qw = [0;l_1+l_2;l_0];

omega1 = [0;0;1];
omega2 = [-1;0;0];
omega3 = [0;1;0];

xi1 = [cross(pb,omega1);omega1]
```

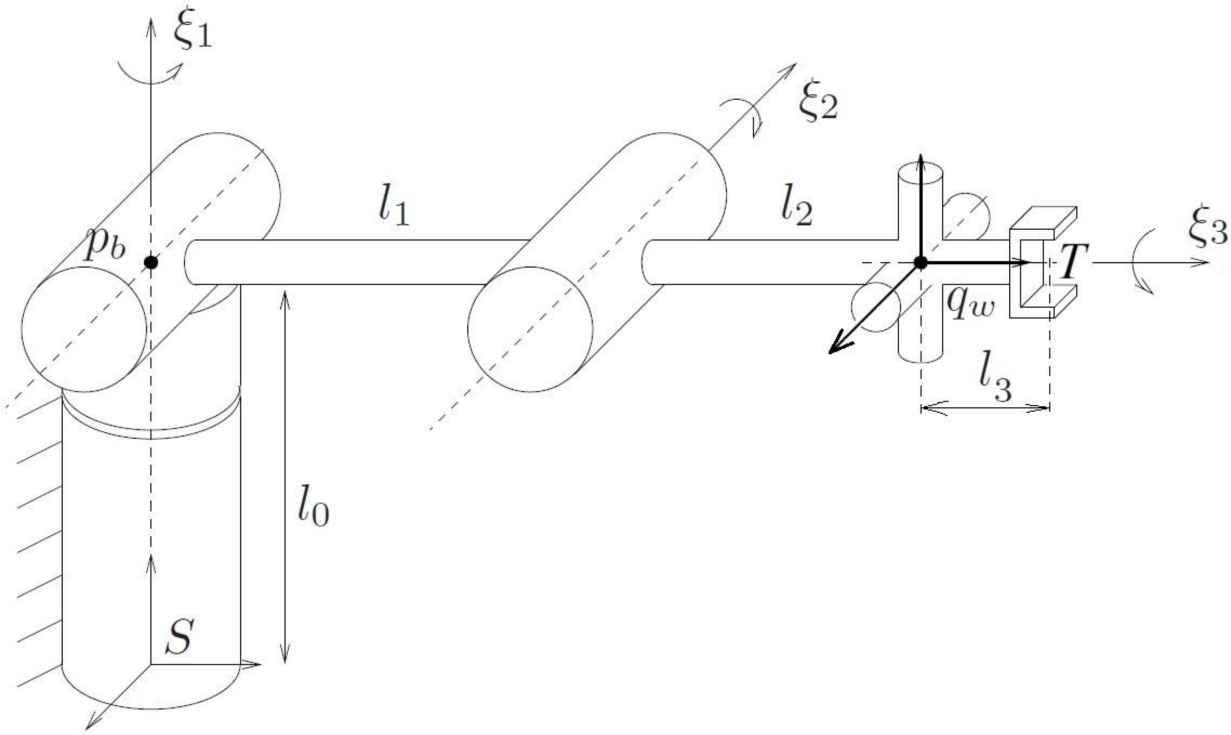
```
xi1 = 6×1
     0
     0
     0
     0
     0
     1
```

```
xi2 = [cross(q2,omega2);omega2]
```

```
xi2 = 6×1
         0
   -0.6500
    0.8000
   -1.0000
         0
         0
```

```
xi3 = [cross(qw,omega3);omega3]
```

```
xi3 = 6×1
   -0.6500
         0
         0
         0
    1.0000
         0
```



Product of exponentials formula, MLS eq (3.3) p. 87:

$$g_{st}(\theta) = e^{\hat{\xi}_1\theta_1}e^{\hat{\xi}_2\theta_2}e^{\hat{\xi}_3\theta_3}g_{st}(0) \ .$$

Therefore,

$$e^{\hat{\xi}_1\theta_1}e^{\hat{\xi}_2\theta_2}e^{\hat{\xi}_3\theta_3} = g_{st}(\theta)g_{st}(0)^{-1} = g_1 \ .$$

```
g1 = g_st/g_st0
```

```
g1 = 4×4
    0.7742   -0.4195    0.4739   -0.4313
   -0.1561    0.5991    0.7853   -0.3344
   -0.6134   -0.6820    0.3983    0.9367
         0         0         0    1.0000
```

Solving problem by applying subproblem 1 three times

If we pick point p on the intersection of axis of ξ_2 and ξ_3, then

$$e^{\hat{\xi}_1\theta_1}e^{\hat{\xi}_2\theta_2}e^{\hat{\xi}_3\theta_3}p = e^{\hat{\xi}_1\theta_1}p = g_1 p \ .$$

From this we can then solve θ_1 using subproblem 1 (MLS p. 99).

```
format long
rr = [pb;1];
pp = [q2;1];
qq = g1*pp;

theta_1 = subproblem_1(xi1,rr,pp,qq)
```

```
theta_1 =
```

```
   0.610865238198015
```

```
format default
rad2deg(theta_1)
```

```
ans =
35
```

If we now pick point p at q_w, then

$$e^{\hat{\xi}_2\theta_2}e^{\hat{\xi}_3\theta_3}p = e^{\hat{\xi}_2\theta_2}p = e^{-\hat{\xi}_1\theta_1}g_1 p \ .$$

From this we can then solve θ_2 using subproblem 1 (MLS p. 99).

```
rr = [q2;1];
pp = [qw;1];
qq = gtwist_num(xi1,-theta_1)*g1*pp;
format long
theta_2 = subproblem_1(xi2,rr,pp,qq)
```

```
theta_2 =
   0.750491578357562
```

```
format default
rad2deg(theta_2)
```

```
ans =
43.0000
```

If we now pick any point p not lying on ξ_3, then

$$e^{\hat{\xi}_3\theta_3}p = e^{-\hat{\xi}_2\theta_2}e^{-\hat{\xi}_1\theta_1}g_1 p \ .$$

From this we can then solve θ_3 using subproblem 1 (MLS p. 99).

```
rr = [qw;1];
pp = [qw+[0.3;0.3;0];1];
qq = gtwist_num(xi2,-theta_2)*gtwist_num(xi1,-theta_1)*g1*pp;
format long
theta_3 = subproblem_1(xi3,rr,pp,qq)
```

```
theta_3 =
   0.994837673636769
```

```
format default
rad2deg(theta_3)
```

```
ans =
57.0000
```

Solving problem with subproblem 2

If we pick point p on the intersection of axis of ξ_2 and ξ_3, then

$$e^{\hat{\xi}_1\theta_1}e^{\hat{\xi}_2\theta_2}e^{\hat{\xi}_3\theta_3}p = e^{\hat{\xi}_1\theta_1}p = g_1 p \ .$$

From this we can then solve θ_1 using subproblem 1 (MLS p. 99), exactly as above.

If we now pick any point p not lying on either ξ_2 or ξ_3, then we can solve $\theta_1$and θ_2 using subproblem 2 (MLS p. 100), as ξ_2 and ξ_3 intersects.

```
rr = [q2;1];
pp = [qw+[0.3;0.3;0];1];
qq = gtwist_num(xi1,-theta_1)*g1*pp;
format long
[theta_2,theta_3] = subproblem_2(xi2,xi3,rr,pp,qq)
```

```
theta_2 = 2×1
   0.750491578357562
   1.359906028027402
theta_3 = 2×1
   0.994837673636768
  -0.994837673636768
```

```
format default
rad2deg(theta_2)
```

```
ans = 2×1
   43.0000
   77.9169
```

```
rad2deg(theta_3)
```

```
ans = 2×1
   57.0000
  -57.0000
```

Testing the 2nd solution

```
thetat2 = theta_2(2);
thetat3 = theta_3(2);

dum = gtwist_num(xi1,theta_1)*gtwist_num(xi2,thetat2)*...
            gtwist_num(xi3,thetat3)*g_st0;
shouldbezeroifvalid4by4 = dum - g_st
```

```
shouldbezeroifvalid4by4 = 4×4
   -0.7985    0.2994   -1.4664    0.2096
    1.1403   -0.4276   -0.8301   -0.2993
    0.7889   -0.2958   -0.2843   -0.2071
         0         0         0         0
```

Clearly the second solution is not valid, as it does not re-generate g_{st} using the product of exponentials formula.

Testing the 1st solution

```
thetat2 = theta_2(1);
thetat3 = theta_3(1);

dum = gtwist_num(xi1,theta_1)*gtwist_num(xi2,thetat2)*...
            gtwist_num(xi3,thetat3)*g_st0;
shouldbezeroifvalid4by4 = dum - g_st
```

```
shouldbezeroifvalid4by4 = 4×4
10^-15 ×
         0         0    0.1665   -0.1110
   -0.3331         0         0         0
```

```
   -0.1110   -0.1110   -0.1665   -0.1110
         0         0         0         0
```

It is concluded that the 1st solution is valid. This solution is in agreement with the solution obtained by applying subproblem 1 thrice.